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NDIC - DATEX II Predefined Location Publication - Predefined Location Set

Release 1.2rev0

Národní Dopravní Informační Centrum

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The specification defines format for publishing predefined location data used to reference status oriented events (travel time, weather station reports, parking, tunnels, etc. accidents etc.).

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NDIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Following terms are used in this document:

format

description of data structure for transferring some kind of information

protocol

description of data transfer process and rules

NDIC

national traffic information centre

DATEX II model

class model for describing information related to the road traffic. The model in form of UML is platform independent.

One of specific platforms for expressing the model is XML.

The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [\[d2ref\]](#). For additional information see [\[d2supp\]](#).

(DATEX II) profile

simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension

extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

B level extension

Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. [16157-1]

C level extension

Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

publication

an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication* [16157-3], *ElaboratedDataPublication* [16157-5], *PredefinedLocationsPublication* [16157-3] etc.

(W3C) XML schema

language, which allows definition of rules for a XML document structure.

XSD

XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri

uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CONCEPTS

Described format allows publishing of predefined locations used for referencing of status information (travel times, level of service, tunnel status, parking occupancy, weather station reports etc.).

2.1 Publication type *PredefinedLocationPublication*

Format is based on DATEX II publication of type *PredefinedLocationPublication* [16157-3]. The profile uses most of data structures, provided by this type of publication and extends it by proprietary location referencing.

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <d2LogicalModel modelBaseVersion="2" xmlns="http://datex2.eu/schema/2/2_0" xmlns:xsi=
  ↳ "http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://datex2.eu/
  ↳ schema/2/2_0 ../schema/schema.xsd">
3   <exchange>
4     <supplierIdentification>
5       <country>cz</country>
6       <nationalIdentifier>NDIC</nationalIdentifier>
7     </supplierIdentification>
8   </exchange>
9   <payloadPublication lang="cs" xsi:type="PredefinedLocationsPublication">
10    <feedType>x-format:cz-ndic_d2-predefined-location-set-v1.2</feedType>
11    <publicationTime>2023-08-15T10:09:22+02:00</publicationTime>
12    <publicationCreator>
13      <country>cz</country>
14      <nationalIdentifier>NDIC</nationalIdentifier>
15    </publicationCreator>
16    <headerInformation>
17      <confidentiality>noRestriction</confidentiality>
18      <informationStatus>real</informationStatus>
19    </headerInformation>
```

2.2 Predefined location description

Each *predefinedLocationContainer* of type *xsi:type="PredefinedLocation"* contains a description of the location of one real world location. There are as many *predefinedLocationContainer* elements in the publication as there are predefined locations. The uniqueness of the predefined locations in this publication is guaranteed by the unique identifier and version.

```
20 <predefinedLocationContainer id="96680652-f696-4b53-b97e-88d16f1943ce" version="1
  ↳ " xsi:type="PredefinedLocation">
```

In a publication, it is possible to use *predefinedLocationContainer* of type *xsi:type="PredefinedNonOrderedLocationGroup"* to describe multiple non-continuous locations together, each separate location is then contained in nested *predefinedLocation* elements.

```
20 <predefinedLocationContainer xsi:type="PredefinedNonOrderedLocationGroup" id=  
↪ "73D74C70-BFA8-408F-86CD-49B3689D9F5D" version="1">  
21 <predefinedLocation id="dde51831-2f37-4f07-8429-286a9d08106c" version="1">
```

The location can be described by a point or a short line segment on the road network. Locations of type area are not used.

The position description methods are further described in the chapter “Location description”.

TIME VALIDITY

The publication start of validity is defined by *publicationTime* for all predefined locations in the publication. No other time record is contained in the publication.

Note: In case of new publication due to changes to predefined location set, old locations remains the same, new locations get new id and updated locations got its version increased.

```
10 <feedType>x-format:cz-ndic_d2-predefined-location-set-v1.2</feedType>  
11 <publicationTime>2023-08-15T10:09:22+02:00</publicationTime>
```


LOCATION REFERENCING

All locations in this publication are predefined and each is contained within element *PrefefinedLocationContainer* globally identified by its identifier and version (these are used to reference the location from other publications). Element *location* (descendant of *NetworkLocation*) with type *Point* or *Linear* contains description of one physical location in one or more location referencing methods.

Three location description methods for a linear location and four location description methods for point location can be used to describe the location. The method use depends on method availability at the location. At least one location description shall be used.

4.1 Location description by means of ALERT-C

The method describes the position using ALERT-C codes, class *alertCLinear* of type *AlertCMethod2Linear* is used. The use of this method depends on the presence of an ALERT-C location at a given location.

The *alertCLinear* method uses a primary location (*alertCMethod2PrimaryPointLocation* element), a secondary location (*alertCMethod2SecondaryPointLocation* element) to describe the event. Of course, the method also contains elements allowing to identify a specific location table.

The primary location is, according to the ALERT-C principle, the place where the accident occurred or where the driver leaves the problem area. The secondary location is the end of the convoy or where the driver first enters the problem area.

In order to find out the direction of the situation, the *alertCDirectionCoded* element is used, which takes the value *positive* if the direction of the traffic flow agrees with the coding of the given route.

Important: WARNING, the meaning of this attribute is therefore exactly the opposite of the *direction* attribute in RDS-TMC, which is used to find the secondary location.

The *alertCLinear* element may occur multiple times in the situation description for long events that must be described by more than one ALERT-C path. This is because ALERT-C only allows locations to be chained within a single path.

Note: The *alertCLinear* method is also used to describe a **point location** (even the inside of an intersection or tunnel, i.e. a short segment). When primary and secondary locations are the same. The direction can be read from the *alertCDirectionCoded* element).

```
111 <location xsi:type="Linear">
112   <alertCLinear xsi:type="AlertCMethod2Linear">
113     <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
114     <alertCLocationTableNumber>25</alertCLocationTableNumber>
115     <alertCLocationTableVersion>9.0</alertCLocationTableVersion>
116     <alertCDirection>
```

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```

117     <alertCDirectionCoded>negative</alertCDirectionCoded>
118   </alertCDirection>
119   <alertCMethod2PrimaryPointLocation>
120     <alertCLocation>
121       <specificLocation>8973</specificLocation>
122     </alertCLocation>
123   </alertCMethod2PrimaryPointLocation>
124   <alertCMethod2SecondaryPointLocation>
125     <alertCLocation>
126       <specificLocation>8974</specificLocation>
127     </alertCLocation>
128   </alertCMethod2SecondaryPointLocation>
129 </alertCLinear>
130 </location>

```

4.2 Location description using mileage

The method uses *linearWithinLinearElement* for the description of the position using mileage. The *roadNumber* element describes the number of the affected road, the *directionRelativeOnLinearSection* element describes the direction of the affected road, and the *fromPoint* and *toPoint* elements describe the distance of the start and end of the event from the start of the road.

```

140 <location xsi:type="Linear">
141   <linearWithinLinearElement>
142     <directionRelativeOnLinearSection>aligned</directionRelativeOnLinearSection>
143     <linearElement>
144       <roadNumber>10</roadNumber>
145     </linearElement>
146     <fromPoint xsi:type="DistanceFromLinearElementStart">
147       <distanceAlong>106797</distanceAlong>
148     </fromPoint>
149     <toPoint xsi:type="DistanceFromLinearElementStart">
150       <distanceAlong>107507</distanceAlong>
151     </toPoint>
152   </linearWithinLinearElement>
153 </location>

```

The method for description of a point event *PointAlongLinearElement* differs by simply describing one distance from the beginning of the identified road instead of two.

```

163 <location xsi:type="Point">
164   <pointAlongLinearElement>
165     <directionRelativeAtPoint>aligned</directionRelativeAtPoint>
166     <linearElement>
167       <roadNumber>D1</roadNumber>
168     </linearElement>
169     <distanceAlongLinearElement xsi:type="DistanceFromLinearElementStart">
170       <distanceAlong>4500</distanceAlong>
171     </distanceAlongLinearElement>
172   </pointAlongLinearElement>
173 </location>

```

4.3 Location description using the Global Network extension

Native method of position input in NDIC using **Extension of DATEX II model of level B**.

The method describes, in the *globalNetworkLinear* element, the location using the segments *linearWithinLinearGNElement* of the Global Network (GN) road network, identified by the *networkVersion* version. Each segment contains a sequence, an identifier referring to the GN, and a relative truncation (for the first and last segment). Furthermore, the position is bounded by the elements *startPoint* and *endPoint* in coordinates (here S-JTSK).

Note: The *globalNetworkLinear* method is also used to describe a **point location** (even the inside of an intersection or tunnel, i.e. a short segment). The *startPoint* and *endPoint* positions contain the same coordinate.

```

27 <location xsi:type="Linear">
28   <linearExtension>
29     <globalNetworkLinear>
30       <networkVersion>22.12</networkVersion>
31       <linearGeometryType>continuous</linearGeometryType>
32       <startPoint>
33         <sjtskPointCoordinates>
34           <sjtskX>-668325.9586937</sjtskX>
35           <sjtskY>-976774.250129909</sjtskY>
36         </sjtskPointCoordinates>
37       </startPoint>
38       <endPoint>
39         <sjtskPointCoordinates>
40           <sjtskX>-668050.419197776</sjtskX>
41           <sjtskY>-976823.636025029</sjtskY>
42         </sjtskPointCoordinates>
43       </endPoint>
44       <linearWithinLinearGNElement>
45         <sectionId>126853</sectionId>
46         <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
47         <orderOfSection>0</orderOfSection>
48         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
49           <percentageDistanceAlong>0.0922403254250082</percentageDistanceAlong>
50         </fromPoint>
51         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
52           <percentageDistanceAlong>0</percentageDistanceAlong>
53         </toPoint>
54       </linearWithinLinearGNElement>
55       <linearWithinLinearGNElement>
56         <sectionId>126599</sectionId>
57         <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
58         <orderOfSection>1</orderOfSection>
59         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
60           <percentageDistanceAlong>1</percentageDistanceAlong>
61         </fromPoint>
62         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
63           <percentageDistanceAlong>0</percentageDistanceAlong>
64         </toPoint>
65       </linearWithinLinearGNElement>
66       <linearWithinLinearGNElement>
67         <sectionId>1461717</sectionId>
68         <directionRelativeOnLinearSection>opposite</

```

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```

69 <directionRelativeOnLinearSection>
70   <orderOfSection>2</orderOfSection>
71   <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
72     <percentageDistanceAlong>1</percentageDistanceAlong>
73   </fromPoint>
74   <toPoint xsi:type="PercentageDistanceAlongLinearElement">
75     <percentageDistanceAlong>0</percentageDistanceAlong>
76   </toPoint>
77 </linearWithinLinearGNElement>
78 <linearWithinLinearGNElement>
79   <sectionId>1461719</sectionId>
80   <directionRelativeOnLinearSection>opposite</
81 <directionRelativeOnLinearSection>
82   <orderOfSection>3</orderOfSection>
83   <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
84     <percentageDistanceAlong>1</percentageDistanceAlong>
85   </fromPoint>
86   <toPoint xsi:type="PercentageDistanceAlongLinearElement">
87     <percentageDistanceAlong>0</percentageDistanceAlong>
88   </toPoint>
89 </linearWithinLinearGNElement>
90 <linearWithinLinearGNElement>
91   <sectionId>2901326</sectionId>
92   <directionRelativeOnLinearSection>opposite</
93 <directionRelativeOnLinearSection>
94   <orderOfSection>4</orderOfSection>
95   <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
96     <percentageDistanceAlong>1</percentageDistanceAlong>
97   </fromPoint>
98   <toPoint xsi:type="PercentageDistanceAlongLinearElement">
99     <percentageDistanceAlong>0.147949455079654</percentageDistanceAlong>
100   </toPoint>
101 </linearWithinLinearGNElement>
</globalNetworkLinear>
</linearExtension>
</location>

```

4.4 Location description using OpenLR extension

The OpenLR method in the *openlrExtendedLinear* element, uses a routing algorithm to describe the location and thus allows on-the-fly localization.

The method uses the reference points *openlrLocationReferencePoint* and *openlrLastLocationReferencePoint* containing coordinates, road direction, and a road type. These reference points are used as the start and end to find the road which is further refined using offsets to the actual extent of the event. In case 2 points are not enough the method allows to use intermediate points.

Note: To describe a **point location** (even the inside of an intersection or tunnel, i.e. a short segment) the *openlrExtendedLinear* method is also used, when offsets refer to a segment with zero length.

```

200 <location xsi:type="Linear">
201   <linearExtension>
202     <openlrExtendedLinear>

```

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```

203     <firstDirection>
204       <openlrLocationReferencePoint>
205         <openlrCoordinate>
206           <latitude>50.765282088427412</latitude>
207           <longitude>15.337521544012871</longitude>
208         </openlrCoordinate>
209         <openlrLineAttributes>
210           <openlrFunctionalRoadClass>FRC1</openlrFunctionalRoadClass>
211           <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
212           <openlrBearing>112</openlrBearing>
213         </openlrLineAttributes>
214         <openlrPathAttributes>
215           <openlrLowestFRCToNextLRPoint>FRC1</openlrLowestFRCToNextLRPoint>
216           <openlrDistanceToNextLRPoint>1240</openlrDistanceToNextLRPoint>
217         </openlrPathAttributes>
218       </openlrLocationReferencePoint>
219       <openlrLastLocationReferencePoint>
220         <openlrCoordinate>
221           <latitude>50.763208691718496</latitude>
222           <longitude>15.347566415800618</longitude>
223         </openlrCoordinate>
224         <openlrLineAttributes>
225           <openlrFunctionalRoadClass>FRC1</openlrFunctionalRoadClass>
226           <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
227           <openlrBearing>283</openlrBearing>
228         </openlrLineAttributes>
229       </openlrLastLocationReferencePoint>
230       <openlrOffsets>
231         <openlrPositiveOffset>120</openlrPositiveOffset>
232         <openlrNegativeOffset>418</openlrNegativeOffset>
233       </openlrOffsets>
234     </firstDirection>
235   </openlrExtendedLinear>
236 </linearExtension>
237 </location>

```

4.5 Location description using Coordinates

The method uses geographic coordinates to describe a situation location in element *pointByCoordinates* using WGS-84 coordinate system.

```

183 <location xsi:type="Point">
184   <pointByCoordinates>
185     <pointCoordinates>
186       <latitude>50.00998</latitude>
187       <longitude>14.54135</longitude>
188     </pointCoordinates>
189   </pointByCoordinates>
190 </location>

```

APPENDIX A: PUBLICATION EXAMPLE

5.1 predefined_location_set-fcd.xml - predefined location set with 3 locations

Publication contains 3 predefined locations.

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
↪instance" xmlns="http://datex2.eu/schema/2/2_0" xsi:schemaLocation="http://datex2.
↪eu/schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication xsi:type="PredefinedLocationsPublication" lang="cs">
    <feedType>x-format:cz-ndic_d2-predefined-location-set-v1.2</feedType>
    <publicationTime>2022-12-01T16:44:19+01:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>ŘSD</nationalIdentifier>
    </publicationCreator>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <predefinedLocationContainer xsi:type="PredefinedLocation" id="TS25760T25761"
↪version="220601">
      <predefinedLocationName>
        <values>
          <value lang="cs">TS25760T25761</value>
        </values>
      </predefinedLocationName>
      <location xsi:type="Linear">
        <alertCLinear xsi:type="AlertCMethod2Linear">
          <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
          <alertCLocationTableNumber>25</alertCLocationTableNumber>
          <alertCLocationTableVersion>9.0</alertCLocationTableVersion>
          <alertCDirection>
            <alertCDirectionCoded>positive</alertCDirectionCoded>
          </alertCDirection>
          <alertCMethod2PrimaryPointLocation>
            <alertCLocation>
```

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```

        <specificLocation>25761</specificLocation>
    </alertCLocation>
</alertCMethod2PrimaryPointLocation>
<alertCMethod2SecondaryPointLocation>
    <alertCLocation>
        <specificLocation>25760</specificLocation>
    </alertCLocation>
</alertCMethod2SecondaryPointLocation>
</alertCLinear>
<linearExtension>
    <openlrExtendedLinear>
        <firstDirection>
            <openlrLocationReferencePoint>
                <openlrCoordinate>
                    <latitude>49.775786944842565</latitude>
                    <longitude>13.347355447257984</longitude>
                </openlrCoordinate>
                <openlrLineAttributes>
                    <openlrFunctionalRoadClass>FRC7</openlrFunctionalRoadClass>
                    <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
                    <openlrBearing>73</openlrBearing>
                </openlrLineAttributes>
                <openlrPathAttributes>
                    <openlrLowestFRCToNextLRPoint>FRC7</openlrLowestFRCToNextLRPoint>
                    <openlrDistanceToNextLRPoint>549</openlrDistanceToNextLRPoint>
                </openlrPathAttributes>
            </openlrLocationReferencePoint>
            <openlrLastLocationReferencePoint>
                <openlrCoordinate>
                    <latitude>49.777293814650236</latitude>
                    <longitude>13.354671919048227</longitude>
                </openlrCoordinate>
                <openlrLineAttributes>
                    <openlrFunctionalRoadClass>FRC7</openlrFunctionalRoadClass>
                    <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
                    <openlrBearing>254</openlrBearing>
                </openlrLineAttributes>
            </openlrLastLocationReferencePoint>
            <openlrOffsets>
                <openlrPositiveOffset>0</openlrPositiveOffset>
                <openlrNegativeOffset>0</openlrNegativeOffset>
            </openlrOffsets>
        </firstDirection>
    </openlrExtendedLinear>
</globalNetworkLinear>
    <networkVersion>22.06</networkVersion>
    <linearGeometryType>continuous</linearGeometryType>
    <linearWithinLinearGNElement>
        <sectionId>3530663</sectionId>
        <directionRelativeOnLinearSection>aligned</
↪directionRelativeOnLinearSection>
        <orderOfSection>0</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
            <percentageDistanceAlong>0</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">

```

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```

        <percentageDistanceAlong>1</percentageDistanceAlong>
      </toPoint>
    </linearWithinLinearGNElement>
    <linearWithinLinearGNElement>
      <sectionId>3530664</sectionId>
      <directionRelativeOnLinearSection>aligned</
↪ directionRelativeOnLinearSection>
      <orderOfSection>1</orderOfSection>
      <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>0</percentageDistanceAlong>
      </fromPoint>
      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>1</percentageDistanceAlong>
      </toPoint>
    </linearWithinLinearGNElement>
    <linearWithinLinearGNElement>
      <sectionId>1375218</sectionId>
      <directionRelativeOnLinearSection>aligned</
↪ directionRelativeOnLinearSection>
      <orderOfSection>2</orderOfSection>
      <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>0</percentageDistanceAlong>
      </fromPoint>
      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>1</percentageDistanceAlong>
      </toPoint>
    </linearWithinLinearGNElement>
    <linearWithinLinearGNElement>
      <sectionId>3850094</sectionId>
      <directionRelativeOnLinearSection>aligned</
↪ directionRelativeOnLinearSection>
      <orderOfSection>3</orderOfSection>
      <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>0</percentageDistanceAlong>
      </fromPoint>
      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>1</percentageDistanceAlong>
      </toPoint>
    </linearWithinLinearGNElement>
    <linearWithinLinearGNElement>
      <sectionId>3850095</sectionId>
      <directionRelativeOnLinearSection>aligned</
↪ directionRelativeOnLinearSection>
      <orderOfSection>4</orderOfSection>
      <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>0</percentageDistanceAlong>
      </fromPoint>
      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>1</percentageDistanceAlong>
      </toPoint>
    </linearWithinLinearGNElement>
    <linearWithinLinearGNElement>
      <sectionId>1846951</sectionId>
      <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
      <orderOfSection>5</orderOfSection>

```

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```

        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>1</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
      <linearWithinLinearGNElement>
        <sectionId>1846952</sectionId>
        <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
        <orderOfSection>6</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>1</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
      <linearWithinLinearGNElement>
        <sectionId>2836873</sectionId>
        <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
        <orderOfSection>7</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>1</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
      <linearWithinLinearGNElement>
        <sectionId>2836874</sectionId>
        <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
        <orderOfSection>8</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>1</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
      <linearWithinLinearGNElement>
        <sectionId>1375231</sectionId>
        <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
        <orderOfSection>9</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>1</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
      <linearWithinLinearGNElement>

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```

        <sectionId>759180</sectionId>
        <directionRelativeOnLinearSection>aligned</
↪directionRelativeOnLinearSection>
        <orderOfSection>10</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
            <percentageDistanceAlong>0</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
            <percentageDistanceAlong>1</percentageDistanceAlong>
        </toPoint>
    </linearWithinLinearGNElement>
    <linearWithinLinearGNElement>
        <sectionId>759181</sectionId>
        <directionRelativeOnLinearSection>aligned</
↪directionRelativeOnLinearSection>
        <orderOfSection>11</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
            <percentageDistanceAlong>0</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
            <percentageDistanceAlong>1</percentageDistanceAlong>
        </toPoint>
    </linearWithinLinearGNElement>
</globalNetworkLinear>
</linearExtension>
</location>
</predefinedLocationContainer>
<predefinedLocationContainer xsi:type="PredefinedLocation" id="TS25761T25760"
↪version="220601">
    <predefinedLocationName>
        <values>
            <value lang="cs">TS25761T25760</value>
        </values>
    </predefinedLocationName>
    <location xsi:type="Linear">
        <alertCLinear xsi:type="AlertCMethod2Linear">
            <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
            <alertCLocationTableNumber>25</alertCLocationTableNumber>
            <alertCLocationTableVersion>9.0</alertCLocationTableVersion>
            <alertCDirection>
                <alertCDirectionCoded>negative</alertCDirectionCoded>
            </alertCDirection>
            <alertCMethod2PrimaryPointLocation>
                <alertCLocation>
                    <specificLocation>25760</specificLocation>
                </alertCLocation>
            </alertCMethod2PrimaryPointLocation>
            <alertCMethod2SecondaryPointLocation>
                <alertCLocation>
                    <specificLocation>25761</specificLocation>
                </alertCLocation>
            </alertCMethod2SecondaryPointLocation>
        </alertCLinear>
    </linearExtension>
    <openlrExtendedLinear>
        <firstDirection>

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```

<openlrLocationReferencePoint>
  <openlrCoordinate>
    <latitude>49.777293814650236</latitude>
    <longitude>13.354671919048227</longitude>
  </openlrCoordinate>
  <openlrLineAttributes>
    <openlrFunctionalRoadClass>FRC7</openlrFunctionalRoadClass>
    <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
    <openlrBearing>254</openlrBearing>
  </openlrLineAttributes>
  <openlrPathAttributes>
    <openlrLowestFRCToNextLRPoint>FRC7</openlrLowestFRCToNextLRPoint>
    <openlrDistanceToNextLRPoint>549</openlrDistanceToNextLRPoint>
  </openlrPathAttributes>
</openlrLocationReferencePoint>
<openlrLastLocationReferencePoint>
  <openlrCoordinate>
    <latitude>49.775786944842565</latitude>
    <longitude>13.347355447257984</longitude>
  </openlrCoordinate>
  <openlrLineAttributes>
    <openlrFunctionalRoadClass>FRC7</openlrFunctionalRoadClass>
    <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
    <openlrBearing>73</openlrBearing>
  </openlrLineAttributes>
</openlrLastLocationReferencePoint>
<openlrOffsets>
  <openlrPositiveOffset>0</openlrPositiveOffset>
  <openlrNegativeOffset>0</openlrNegativeOffset>
</openlrOffsets>
</firstDirection>
</openlrExtendedLinear>
<globalNetworkLinear>
  <networkVersion>22.06</networkVersion>
  <linearGeometryType>continuous</linearGeometryType>
  <linearWithinLinearGNElement>
    <sectionId>759181</sectionId>
    <directionRelativeOnLinearSection>opposite</
->directionRelativeOnLinearSection>
    <orderOfSection>0</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>1</percentageDistanceAlong>
    </fromPoint>
    <toPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>0</percentageDistanceAlong>
    </toPoint>
  </linearWithinLinearGNElement>
  <linearWithinLinearGNElement>
    <sectionId>759180</sectionId>
    <directionRelativeOnLinearSection>opposite</
->directionRelativeOnLinearSection>
    <orderOfSection>1</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>1</percentageDistanceAlong>
    </fromPoint>
    <toPoint xsi:type="PercentageDistanceAlongLinearElement">

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```

        <percentageDistanceAlong>0</percentageDistanceAlong>
      </toPoint>
    </linearWithinLinearGNElement>
    <linearWithinLinearGNElement>
      <sectionId>1375231</sectionId>
      <directionRelativeOnLinearSection>aligned</
↪directionRelativeOnLinearSection>
      <orderOfSection>2</orderOfSection>
      <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>0</percentageDistanceAlong>
      </fromPoint>
      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>1</percentageDistanceAlong>
      </toPoint>
    </linearWithinLinearGNElement>
    <linearWithinLinearGNElement>
      <sectionId>2836874</sectionId>
      <directionRelativeOnLinearSection>aligned</
↪directionRelativeOnLinearSection>
      <orderOfSection>3</orderOfSection>
      <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>0</percentageDistanceAlong>
      </fromPoint>
      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>1</percentageDistanceAlong>
      </toPoint>
    </linearWithinLinearGNElement>
    <linearWithinLinearGNElement>
      <sectionId>2836873</sectionId>
      <directionRelativeOnLinearSection>aligned</
↪directionRelativeOnLinearSection>
      <orderOfSection>4</orderOfSection>
      <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>0</percentageDistanceAlong>
      </fromPoint>
      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>1</percentageDistanceAlong>
      </toPoint>
    </linearWithinLinearGNElement>
    <linearWithinLinearGNElement>
      <sectionId>1846952</sectionId>
      <directionRelativeOnLinearSection>aligned</
↪directionRelativeOnLinearSection>
      <orderOfSection>5</orderOfSection>
      <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>0</percentageDistanceAlong>
      </fromPoint>
      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>1</percentageDistanceAlong>
      </toPoint>
    </linearWithinLinearGNElement>
    <linearWithinLinearGNElement>
      <sectionId>1846951</sectionId>
      <directionRelativeOnLinearSection>aligned</
↪directionRelativeOnLinearSection>
      <orderOfSection>6</orderOfSection>

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```

        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>1</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
      <linearWithinLinearGNElement>
        <sectionId>3850095</sectionId>
        <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
        <orderOfSection>7</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>1</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
      <linearWithinLinearGNElement>
        <sectionId>3850094</sectionId>
        <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
        <orderOfSection>8</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>1</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
      <linearWithinLinearGNElement>
        <sectionId>1375218</sectionId>
        <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
        <orderOfSection>9</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>1</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
      <linearWithinLinearGNElement>
        <sectionId>3530664</sectionId>
        <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
        <orderOfSection>10</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>1</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
          <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
      </linearWithinLinearGNElement>
      <linearWithinLinearGNElement>

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```

        <sectionId>3530663</sectionId>
        <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
        <orderOfSection>11</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
            <percentageDistanceAlong>1</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
            <percentageDistanceAlong>0</percentageDistanceAlong>
        </toPoint>
        </linearWithinLinearGNElement>
        </globalNetworkLinear>
        </linearExtension>
        </location>
        </predefinedLocationContainer>
        <predefinedLocationContainer xsi:type="PredefinedLocation" id="TS27072T27073"
↪version="220601">
            <predefinedLocationName>
                <values>
                    <value lang="cs">TS27072T27073</value>
                </values>
            </predefinedLocationName>
            <location xsi:type="Linear">
                <alertCLinear xsi:type="AlertCMethod2Linear">
                    <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
                    <alertCLocationTableNumber>25</alertCLocationTableNumber>
                    <alertCLocationTableVersion>9.0</alertCLocationTableVersion>
                    <alertCDirection>
                        <alertCDirectionCoded>positive</alertCDirectionCoded>
                    </alertCDirection>
                    <alertCMethod2PrimaryPointLocation>
                        <alertCLocation>
                            <specificLocation>27073</specificLocation>
                        </alertCLocation>
                    </alertCMethod2PrimaryPointLocation>
                    <alertCMethod2SecondaryPointLocation>
                        <alertCLocation>
                            <specificLocation>27072</specificLocation>
                        </alertCLocation>
                    </alertCMethod2SecondaryPointLocation>
                </alertCLinear>
            </linearExtension>
            <openlrExtendedLinear>
                <firstDirection>
                    <openlrLocationReferencePoint>
                        <openlrCoordinate>
                            <latitude>49.844782274277328</latitude>
                            <longitude>18.353979148213046</longitude>
                        </openlrCoordinate>
                        <openlrLineAttributes>
                            <openlrFunctionalRoadClass>FRC6</openlrFunctionalRoadClass>
                            <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
                            <openlrBearing>227</openlrBearing>
                        </openlrLineAttributes>
                        <openlrPathAttributes>
                            <openlrLowestFRCToNextLRPoint>FRC6</openlrLowestFRCToNextLRPoint>

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```

        <openlrDistanceToNextLRPoint>287</openlrDistanceToNextLRPoint>
      </openlrPathAttributes>
    </openlrLocationReferencePoint>
    <openlrLastLocationReferencePoint>
      <openlrCoordinate>
        <latitude>49.843366179913971</latitude>
        <longitude>18.350653805486978</longitude>
      </openlrCoordinate>
      <openlrLineAttributes>
        <openlrFunctionalRoadClass>FRC6</openlrFunctionalRoadClass>
        <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
        <openlrBearing>78</openlrBearing>
      </openlrLineAttributes>
    </openlrLastLocationReferencePoint>
    <openlrOffsets>
      <openlrPositiveOffset>0</openlrPositiveOffset>
      <openlrNegativeOffset>0</openlrNegativeOffset>
    </openlrOffsets>
  </firstDirection>
</openlrExtendedLinear>
<globalNetworkLinear>
  <networkVersion>22.06</networkVersion>
  <linearGeometryType>continuous</linearGeometryType>
  <linearWithinLinearGNElement>
    <sectionId>461235</sectionId>
    <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
    <orderOfSection>0</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>1</percentageDistanceAlong>
    </fromPoint>
    <toPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>0</percentageDistanceAlong>
    </toPoint>
  </linearWithinLinearGNElement>
  <linearWithinLinearGNElement>
    <sectionId>461471</sectionId>
    <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
    <orderOfSection>1</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>1</percentageDistanceAlong>
    </fromPoint>
    <toPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>0</percentageDistanceAlong>
    </toPoint>
  </linearWithinLinearGNElement>
  <linearWithinLinearGNElement>
    <sectionId>461547</sectionId>
    <directionRelativeOnLinearSection>opposite</
↪directionRelativeOnLinearSection>
    <orderOfSection>2</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>1</percentageDistanceAlong>
    </fromPoint>
    <toPoint xsi:type="PercentageDistanceAlongLinearElement">

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```

        <percentageDistanceAlong>0</percentageDistanceAlong>
      </toPoint>
    </linearWithinLinearGNElement>
    <linearWithinLinearGNElement>
      <sectionId>461605</sectionId>
      <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
      <orderOfSection>3</orderOfSection>
      <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>1</percentageDistanceAlong>
      </fromPoint>
      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
        <percentageDistanceAlong>0</percentageDistanceAlong>
      </toPoint>
    </linearWithinLinearGNElement>
  </globalNetworkLinear>
</linearExtension>
</location>
</predefinedLocationContainer>
</payloadPublication>
</d2LogicalModel>

```

5.1.1 Root element and publication envelope

Root element *d2LogicalModel* declares, that this message is using DATEX II.

Element *exchange* contains provider identifier.

Element *payloadPublication* is an envelope for actual information payload containing general information and one or more elements *predefinedLocationContainer*.

Attribute *xsi:type="PredefinedLocationsPublication"* in element *payloadPublication* determines, that this envelope is of type predefined location publication.

Element *feedType* states publication schema by its uri.

Element *publicationTime* states publication time of this publication as a whole.

Element *publicationCreator* identifies creator of the publication.

5.1.2 Description of predefined locations

The predefined location is described in the *predefinedLocationContainer* element by a directly nested description.

The example publication describes 3 predefined locations with the identifiers *id="TS25760T25761"*, *id="TS25761T25760"* and *id="TS27072T27073"*. All locations contain 3 methods describing the same slice of reality. These are sections of the FCD data location network.

Position description methods

The *predefinedLocationName* element contains the name of the location, in this case the same as the identifier.

The *location* element then contains the location description itself. The *xsi:type="Linear"* attribute indicates that this is a linear type of location.

In the example, there are 3 location descriptions used for all *predefinedLocationContainer*:

- tmc location using location tables in the *alertCLinear* element of type *AlertCMethod2Linear*,
- description using references to Global Network section ids in the *globalNetworkLinear* extension, and
- description using the OpenLR routing method in the *openlrExtendedLinear* extension.

APPENDIX B: W3C XML SCHEMA

XML structure of a transmitted message is defined by following W3C XML schema:

6.1 *schema.xsd*

```

1 <?xml version="1.0" encoding="utf-8" standalone="no"?>
2 <xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified"
  ↳ xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace=
  ↳ "http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
3   <xs:complexType name="_ExtensionType">
4     <xs:sequence>
5       <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
6     </xs:sequence>
7   </xs:complexType>
8   <xs:complexType name="_IntermediatePointOnLinearElement">
9     <xs:sequence>
10      <xs:element name="referent" type="D2LogicalModel:Referent" minOccurs="1"
  ↳ maxOccurs="1" />
11    </xs:sequence>
12    <xs:attribute name="index" type="xs:int" use="required" />
13  </xs:complexType>
14  <xs:complexType name="_LinearExtensionType">
15    <xs:sequence>
16      <xs:element name="openlrExtendedLinear" type=
  ↳ "D2LogicalModel:OpenlrExtendedLinear" minOccurs="0" />
17      <xs:element name="globalNetworkLinear" type="D2LogicalModel:GlobalNetworkLinear"
  ↳ minOccurs="0" />
18      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
19    </xs:sequence>
20  </xs:complexType>
21  <xs:complexType name="_PointExtensionType">
22    <xs:sequence>
23      <xs:element name="pointWithinLinearGNElement" type=
  ↳ "D2LogicalModel:PointWithinLinearGNElement" minOccurs="0" />
24      <xs:element name="openlrExtendedPoint" type="D2LogicalModel:OpenlrExtendedPoint"
  ↳ minOccurs="0" />
25      <xs:element name="pointByMultiCoordinates" type=
  ↳ "D2LogicalModel:PointByMultiCoordinates" minOccurs="0" />
26      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
27    </xs:sequence>
28  </xs:complexType>

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```

29 <xs:complexType name="_PredefinedLocationVersionedReference">
30   <xs:complexContent>
31     <xs:extension base="D2LogicalModel:VersionedReference">
32       <xs:attribute name="targetClass" use="required" fixed="PredefinedLocation" />
33     </xs:extension>
34   </xs:complexContent>
35 </xs:complexType>
36 <xs:complexType name="AlertCDirection">
37   <xs:sequence>
38     <xs:element name="alertCDirectionCoded" type="D2LogicalModel:AlertCDirectionEnum"
39     ↪ " minOccurs="1" maxOccurs="1" />
40     <xs:element name="alertCDirectionNamed" type="D2LogicalModel:MultilingualString"
41     ↪ " minOccurs="0" maxOccurs="1" />
42     <xs:element name="alertCDirectionSense" type="D2LogicalModel:Boolean" minOccurs=
43     ↪ "0" maxOccurs="1" />
44     <xs:element name="alertCDirectionExtension" type="D2LogicalModel:_ExtensionType"
45     ↪ " minOccurs="0" />
46   </xs:sequence>
47 </xs:complexType>
48 <xs:simpleType name="AlertCDirectionEnum">
49   <xs:restriction base="xs:string">
50     <xs:enumeration value="both" />
51     <xs:enumeration value="negative" />
52     <xs:enumeration value="positive" />
53     <xs:enumeration value="unknown" />
54   </xs:restriction>
55 </xs:simpleType>
56 <xs:complexType name="AlertCLinear" abstract="true">
57   <xs:sequence>
58     <xs:element name="alertCLocationCountryCode" type="D2LogicalModel:String"
59     ↪ minOccurs="1" maxOccurs="1" />
60     <xs:element name="alertCLocationTableNumber" type="D2LogicalModel:String"
61     ↪ minOccurs="1" maxOccurs="1" />
62     <xs:element name="alertCLocationTableVersion" type="D2LogicalModel:String"
63     ↪ minOccurs="1" maxOccurs="1" />
64     <xs:element name="alertCLinearExtension" type="D2LogicalModel:_ExtensionType"
65     ↪ minOccurs="0" />
66   </xs:sequence>
67 </xs:complexType>
68 <xs:complexType name="AlertCLinearByCode">
69   <xs:complexContent>
70     <xs:extension base="D2LogicalModel:AlertCLinear">
71       <xs:sequence>
72         <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
73         <xs:element name="locationCodeForLinearLocation" type=
74         ↪ "D2LogicalModel:AlertCLocation" />
75         <xs:element name="alertCLinearByCodeExtension" type="D2LogicalModel:_
76         ↪ ExtensionType" minOccurs="0" />
77       </xs:sequence>
78     </xs:extension>
79   </xs:complexContent>
80 </xs:complexType>
81 <xs:complexType name="AlertCLocation">
82   <xs:sequence>
83     <xs:element name="alertCLocationName" type="D2LogicalModel:MultilingualString"
84     ↪ minOccurs="0" maxOccurs="1" />

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```

74     <xs:element name="specificLocation" type="D2LogicalModel:AlertCLocationCode"
↪ minOccurs="1" maxOccurs="1" />
75     <xs:element name="alertCLocationExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
76     </xs:sequence>
77 </xs:complexType>
78 <xs:simpleType name="AlertCLocationCode">
79     <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
80 </xs:simpleType>
81 <xs:complexType name="AlertCMethod2Linear">
82     <xs:complexContent>
83         <xs:extension base="D2LogicalModel:AlertCLinear">
84             <xs:sequence>
85                 <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
86                 <xs:element name="alertCMethod2PrimaryPointLocation" type=
↪ "D2LogicalModel:AlertCMethod2PrimaryPointLocation" />
87                 <xs:element name="alertCMethod2SecondaryPointLocation" type=
↪ "D2LogicalModel:AlertCMethod2SecondaryPointLocation" />
88                 <xs:element name="alertCMethod2LinearExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
89             </xs:sequence>
90         </xs:extension>
91     </xs:complexContent>
92 </xs:complexType>
93 <xs:complexType name="AlertCMethod2Point">
94     <xs:complexContent>
95         <xs:extension base="D2LogicalModel:AlertCPoint">
96             <xs:sequence>
97                 <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
98                 <xs:element name="alertCMethod2PrimaryPointLocation" type=
↪ "D2LogicalModel:AlertCMethod2PrimaryPointLocation" />
99                 <xs:element name="alertCMethod2PointExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
100             </xs:sequence>
101         </xs:extension>
102     </xs:complexContent>
103 </xs:complexType>
104 <xs:complexType name="AlertCMethod2PrimaryPointLocation">
105     <xs:sequence>
106         <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" />
107         <xs:element name="alertCMethod2PrimaryPointLocationExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
108     </xs:sequence>
109 </xs:complexType>
110 <xs:complexType name="AlertCMethod2SecondaryPointLocation">
111     <xs:sequence>
112         <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" />
113         <xs:element name="alertCMethod2SecondaryPointLocationExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
114     </xs:sequence>
115 </xs:complexType>
116 <xs:complexType name="AlertCPoint" abstract="true">
117     <xs:sequence>
118         <xs:element name="alertCLocationCountryCode" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
119         <xs:element name="alertCLocationTableNumber" type="D2LogicalModel:String"

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```

120 <minOccurs="1" maxOccurs="1" />
    <xs:element name="alertCLocationTableVersion" type="D2LogicalModel:String"
121 <minOccurs="1" maxOccurs="1" />
    <xs:element name="alertCPointExtension" type="D2LogicalModel:_ExtensionType"
    <minOccurs="0" />
122 </xs:sequence>
123 </xs:complexType>
124 <xs:simpleType name="AngleInDegrees">
125   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
126 </xs:simpleType>
127 <xs:simpleType name="AreaOfInterestEnum">
128   <xs:restriction base="xs:string">
129     <xs:enumeration value="continentWide" />
130     <xs:enumeration value="national" />
131     <xs:enumeration value="neighbouringCountries" />
132     <xs:enumeration value="notSpecified" />
133     <xs:enumeration value="regional" />
134   </xs:restriction>
135 </xs:simpleType>
136 <xs:simpleType name="Boolean">
137   <xs:restriction base="xs:boolean" />
138 </xs:simpleType>
139 <xs:simpleType name="ConfidentialityValueEnum">
140   <xs:restriction base="xs:string">
141     <xs:enumeration value="internalUse" />
142     <xs:enumeration value="noRestriction" />
143     <xs:enumeration value="restrictedToAuthorities" />
144     <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators" />
145     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers" />
146     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms" />
147   </xs:restriction>
148 </xs:simpleType>
149 <xs:simpleType name="CountryEnum">
150   <xs:restriction base="xs:string">
151     <xs:enumeration value="at" />
152     <xs:enumeration value="be" />
153     <xs:enumeration value="bg" />
154     <xs:enumeration value="ch" />
155     <xs:enumeration value="cs" />
156     <xs:enumeration value="cy" />
157     <xs:enumeration value="cz" />
158     <xs:enumeration value="de" />
159     <xs:enumeration value="dk" />
160     <xs:enumeration value="ee" />
161     <xs:enumeration value="es" />
162     <xs:enumeration value="fi" />
163     <xs:enumeration value="fo" />
164     <xs:enumeration value="fr" />
165     <xs:enumeration value="gb" />
166     <xs:enumeration value="gg" />
167     <xs:enumeration value="gi" />
168     <xs:enumeration value="gr" />
169     <xs:enumeration value="hr" />
170     <xs:enumeration value="hu" />
171     <xs:enumeration value="ie" />
172     <xs:enumeration value="im" />

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173     <xs:enumeration value="is" />
174     <xs:enumeration value="it" />
175     <xs:enumeration value="je" />
176     <xs:enumeration value="li" />
177     <xs:enumeration value="lt" />
178     <xs:enumeration value="lu" />
179     <xs:enumeration value="lv" />
180     <xs:enumeration value="ma" />
181     <xs:enumeration value="mc" />
182     <xs:enumeration value="mk" />
183     <xs:enumeration value="mt" />
184     <xs:enumeration value="nl" />
185     <xs:enumeration value="no" />
186     <xs:enumeration value="pl" />
187     <xs:enumeration value="pt" />
188     <xs:enumeration value="ro" />
189     <xs:enumeration value="se" />
190     <xs:enumeration value="si" />
191     <xs:enumeration value="sk" />
192     <xs:enumeration value="sm" />
193     <xs:enumeration value="tr" />
194     <xs:enumeration value="va" />
195     <xs:enumeration value="other" />
196 </xs:restriction>
197 </xs:simpleType>
198 <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel">
199     <xs:unique name="_d2LogicalModelPredefinedNonOrderedLocationGroupConstraint">
200         <xs:selector xpath="."/D2LogicalModel:predefinedNonOrderedLocationGroup" />
201         <xs:field xpath="@id" />
202         <xs:field xpath="@version" />
203     </xs:unique>
204     <xs:unique name="_d2LogicalModelPredefinedLocationConstraint">
205         <xs:selector xpath="."/D2LogicalModel:predefinedLocation" />
206         <xs:field xpath="@id" />
207         <xs:field xpath="@version" />
208     </xs:unique>
209 </xs:element>
210 <xs:complexType name="D2LogicalModel">
211     <xs:sequence>
212         <xs:element name="exchange" type="D2LogicalModel:Exchange" />
213         <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication"
214             ↪ minOccurs="0" />
215         <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_ExtensionType"
216             ↪ minOccurs="0" />
217     </xs:sequence>
218     <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
219 </xs:complexType>
220 <xs:simpleType name="DateTime">
221     <xs:restriction base="xs:dateTime" />
222 </xs:simpleType>
223 <xs:simpleType name="DirectionEnum">
224     <xs:restriction base="xs:string">
225         <xs:enumeration value="allDirections" />
226         <xs:enumeration value="bothWays" />
227         <xs:enumeration value="clockwise" />
228         <xs:enumeration value="anticlockwise" />

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227     <xs:enumeration value="innerRing" />
228     <xs:enumeration value="outerRing" />
229     <xs:enumeration value="northBound" />
230     <xs:enumeration value="northEastBound" />
231     <xs:enumeration value="eastBound" />
232     <xs:enumeration value="southEastBound" />
233     <xs:enumeration value="southBound" />
234     <xs:enumeration value="southWestBound" />
235     <xs:enumeration value="westBound" />
236     <xs:enumeration value="northWestBound" />
237     <xs:enumeration value="inboundTowardsTown" />
238     <xs:enumeration value="outboundFromTown" />
239     <xs:enumeration value="unknown" />
240     <xs:enumeration value="opposite" />
241     <xs:enumeration value="other" />
242   </xs:restriction>
243 </xs:simpleType>
244 <xs:complexType name="DistanceAlongLinearElement" abstract="true">
245   <xs:sequence>
246     <xs:element name="distanceAlongLinearElementExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
247   </xs:sequence>
248 </xs:complexType>
249 <xs:complexType name="DistanceFromLinearElementReferent">
250   <xs:complexContent>
251     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
252       <xs:sequence>
253         <xs:element name="distanceAlong" type="D2LogicalModel:MetresAsFloat" ↪
↪ minOccurs="1" maxOccurs="1" />
254         <xs:element name="fromReferent" type="D2LogicalModel:Referent" />
255         <xs:element name="towardsReferent" type="D2LogicalModel:Referent" minOccurs=
↪ "0" />
256         <xs:element name="distanceFromLinearElementReferentExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
257       </xs:sequence>
258     </xs:extension>
259   </xs:complexContent>
260 </xs:complexType>
261 <xs:complexType name="DistanceFromLinearElementStart">
262   <xs:complexContent>
263     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
264       <xs:sequence>
265         <xs:element name="distanceAlong" type="D2LogicalModel:MetresAsFloat" ↪
↪ minOccurs="1" maxOccurs="1" />
266         <xs:element name="distanceFromLinearElementStartExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
267       </xs:sequence>
268     </xs:extension>
269   </xs:complexContent>
270 </xs:complexType>
271 <xs:complexType name="Exchange">
272   <xs:sequence>
273     <xs:element name="supplierIdentification" type=
↪ "D2LogicalModel:InternationalIdentifier" />
274     <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType" ↪
↪ minOccurs="0" />

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275     </xs:sequence>
276 </xs:complexType>
277 <xs:simpleType name="Float">
278   <xs:restriction base="xs:float" />
279 </xs:simpleType>
280 <xs:simpleType name="GeometryTypeEnum">
281   <xs:restriction base="xs:string">
282     <xs:enumeration value="continuous" />
283     <xs:enumeration value="discontinuous" />
284   </xs:restriction>
285 </xs:simpleType>
286 <xs:complexType name="GlobalNetworkLinear">
287   <xs:sequence>
288     <xs:element name="networkVersion" type="D2LogicalModel:String" minOccurs="1"
↪ maxOccurs="1" />
289     <xs:element name="linearGeometryType" type="D2LogicalModel:GeometryTypeEnum"
↪ minOccurs="1" maxOccurs="1" />
290     <xs:element name="startPoint" type="D2LogicalModel:PointByMultiCoordinates"
↪ minOccurs="0" />
291     <xs:element name="endPoint" type="D2LogicalModel:PointByMultiCoordinates"
↪ minOccurs="0" />
292     <xs:element name="linearWithinLinearGNElement" type=
↪ "D2LogicalModel:LinearWithinLinearGNElement" maxOccurs="unbounded" />
293   </xs:sequence>
294 </xs:complexType>
295 <xs:complexType name="GroupOfLocations" abstract="true">
296   <xs:sequence>
297     <xs:element name="groupOfLocationsExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
298   </xs:sequence>
299 </xs:complexType>
300 <xs:complexType name="HeaderInformation">
301   <xs:sequence>
302     <xs:element name="areaOfInterest" type="D2LogicalModel:AreaOfInterestEnum"
↪ minOccurs="0" maxOccurs="1" />
303     <xs:element name="confidentiality" type="D2LogicalModel:ConfidentialityValueEnum"
↪ minOccurs="1" maxOccurs="1" />
304     <xs:element name="informationStatus" type="D2LogicalModel:InformationStatusEnum"
↪ minOccurs="1" maxOccurs="1" />
305     <xs:element name="urgency" type="D2LogicalModel:UrgencyEnum" minOccurs="0"
↪ maxOccurs="1" />
306     <xs:element name="headerInformationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
307   </xs:sequence>
308 </xs:complexType>
309 <xs:simpleType name="HeightGradeEnum">
310   <xs:restriction base="xs:string">
311     <xs:enumeration value="aboveGrade" />
312     <xs:enumeration value="atGrade" />
313     <xs:enumeration value="belowGrade" />
314   </xs:restriction>
315 </xs:simpleType>
316 <xs:simpleType name="InformationStatusEnum">
317   <xs:restriction base="xs:string">
318     <xs:enumeration value="real" />
319     <xs:enumeration value="securityExercise" />

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320     <xs:enumeration value="technicalExercise" />
321     <xs:enumeration value="test" />
322   </xs:restriction>
323 </xs:simpleType>
324 <xs:complexType name="InternationalIdentifier">
325   <xs:sequence>
326     <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1"
↪ maxOccurs="1" />
327     <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs="1"
↪ " maxOccurs="1" />
328     <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
329   </xs:sequence>
330 </xs:complexType>
331 <xs:simpleType name="Language">
332   <xs:restriction base="xs:language" />
333 </xs:simpleType>
334 <xs:complexType name="Linear">
335   <xs:complexContent>
336     <xs:extension base="D2LogicalModel:NetworkLocation">
337       <xs:sequence>
338         <xs:element name="alertCLinear" type="D2LogicalModel:AlertCLinear"
↪ minOccurs="0" />
339         <xs:element name="linearWithinLinearElement" type=
↪ "D2LogicalModel:LinearWithinLinearElement" minOccurs="0" />
340         <xs:element name="linearExtension" type="D2LogicalModel:_LinearExtensionType
↪ " minOccurs="0" />
341       </xs:sequence>
342     </xs:extension>
343   </xs:complexContent>
344 </xs:complexType>
345 <xs:complexType name="LinearElement">
346   <xs:sequence>
347     <xs:element name="roadName" type="D2LogicalModel:MultilingualString" minOccurs=
↪ "0" maxOccurs="1" />
348     <xs:element name="roadNumber" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
349     <xs:element name="linearElementReferenceModel" type="D2LogicalModel:String"
↪ minOccurs="0" maxOccurs="1" />
350     <xs:element name="linearElementReferenceModelVersion" type=
↪ "D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
351     <xs:element name="linearElementNature" type=
↪ "D2LogicalModel:LinearElementNatureEnum" minOccurs="0" maxOccurs="1" />
352     <xs:element name="linearElementExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
353   </xs:sequence>
354 </xs:complexType>
355 <xs:complexType name="LinearElementByCode">
356   <xs:complexContent>
357     <xs:extension base="D2LogicalModel:LinearElement">
358       <xs:sequence>
359         <xs:element name="linearElementIdentifier" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
360         <xs:element name="linearElementByCodeExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
361       </xs:sequence>

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362     </xs:extension>
363   </xs:complexContent>
364 </xs:complexType>
365 <xs:complexType name="LinearElementByPoints">
366   <xs:complexContent>
367     <xs:extension base="D2LogicalModel:LinearElement">
368       <xs:sequence>
369         <xs:element name="startPointOfLinearElement" type="D2LogicalModel:Referent" />
370       </xs:sequence>
371       <xs:element name="intermediatePointOnLinearElement" type="D2LogicalModel:_
372       <IntermediatePointOnLinearElement" minOccurs="0" maxOccurs="unbounded" />
373       <xs:element name="endPointOfLinearElement" type="D2LogicalModel:Referent" />
374       <xs:element name="linearElementByPointsExtension" type="D2LogicalModel:_
375       <ExtensionType" minOccurs="0" />
376     </xs:sequence>
377   </xs:extension>
378 </xs:complexContent>
379 </xs:complexType>
380 <xs:simpleType name="LinearElementNatureEnum">
381   <xs:restriction base="xs:string">
382     <xs:enumeration value="road" />
383     <xs:enumeration value="roadSection" />
384     <xs:enumeration value="slipRoad" />
385     <xs:enumeration value="other" />
386   </xs:restriction>
387 </xs:simpleType>
388 <xs:simpleType name="LinearReferencingDirectionEnum">
389   <xs:restriction base="xs:string">
390     <xs:enumeration value="both" />
391     <xs:enumeration value="opposite" />
392     <xs:enumeration value="aligned" />
393     <xs:enumeration value="unknown" />
394   </xs:restriction>
395 </xs:simpleType>
396 <xs:complexType name="LinearWithinLinearElement">
397   <xs:sequence>
398     <xs:element name="administrativeAreaOfLinearSection" type=
399     <"D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
400     <xs:element name="directionBoundOnLinearSection" type=
401     <"D2LogicalModel:DirectionEnum" minOccurs="0" maxOccurs="1" />
402     <xs:element name="directionRelativeOnLinearSection" type=
403     <"D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="0" maxOccurs="1" />
404     <xs:element name="heightGradeOfLinearSection" type=
405     <"D2LogicalModel:HeightGradeEnum" minOccurs="0" maxOccurs="1" />
406     <xs:element name="linearElement" type="D2LogicalModel:LinearElement" />
407     <xs:element name="fromPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
408     <xs:element name="toPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
409     <xs:element name="linearWithinLinearElementExtension" type="D2LogicalModel:_
410     <ExtensionType" minOccurs="0" />
411   </xs:sequence>
412 </xs:complexType>
413 <xs:complexType name="LinearWithinLinearGNElement">
414   <xs:sequence>
415     <xs:element name="sectionId" type="D2LogicalModel:NonNegativeInteger" minOccurs=
416     <"0" maxOccurs="1" />
417     <xs:element name="directionRelativeOnLinearSection" type=

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409 <=>"D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="1" maxOccurs="1" />
    <xs:element name="orderOfSection" type="D2LogicalModel:NonNegativeInteger"
410 <=>minOccurs="0" maxOccurs="1" />
    <xs:element name="fromPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
411 <xs:element name="toPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
412 <xs:element name="linearWithinLinearGNElementExtension" type="D2LogicalModel:_
    <=>ExtensionType" minOccurs="0" />
413 </xs:sequence>
414 </xs:complexType>
415 <xs:complexType name="Location" abstract="true">
416 <xs:complexContent>
417 <xs:extension base="D2LogicalModel:GroupOfLocations">
418 <xs:sequence>
419 <xs:element name="locationExtension" type="D2LogicalModel:_ExtensionType"
    <=>minOccurs="0" />
420 </xs:sequence>
421 </xs:extension>
422 </xs:complexContent>
423 </xs:complexType>
424 <xs:complexType name="LocationByReference">
425 <xs:complexContent>
426 <xs:extension base="D2LogicalModel:Location">
427 <xs:sequence>
428 <xs:element name="predefinedLocationReference" type="D2LogicalModel:_
    <=>PredefinedLocationVersionedReference" minOccurs="1" maxOccurs="1" />
429 <xs:element name="locationByReferenceExtension" type="D2LogicalModel:_
    <=>ExtensionType" minOccurs="0" />
430 </xs:sequence>
431 </xs:extension>
432 </xs:complexContent>
433 </xs:complexType>
434 <xs:simpleType name="MetresAsFloat">
435 <xs:restriction base="D2LogicalModel:Float" />
436 </xs:simpleType>
437 <xs:simpleType name="MetresAsNonNegativeInteger">
438 <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
439 </xs:simpleType>
440 <xs:complexType name="MultilingualString">
441 <xs:sequence>
442 <xs:element name="values">
443 <xs:complexType>
444 <xs:sequence>
445 <xs:element name="value" type="D2LogicalModel:MultilingualStringValue"
    <=>maxOccurs="unbounded" />
446 </xs:sequence>
447 </xs:complexType>
448 </xs:element>
449 </xs:sequence>
450 </xs:complexType>
451 <xs:complexType name="MultilingualStringValue">
452 <xs:simpleContent>
453 <xs:extension base="D2LogicalModel:MultilingualStringValueType">
454 <xs:attribute name="lang" type="xs:language" />
455 </xs:extension>
456 </xs:simpleContent>
457 </xs:complexType>

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458 <xs:simpleType name="MultilingualStringValue">
459   <xs:restriction base="xs:string">
460     <xs:maxLength value="1024" />
461   </xs:restriction>
462 </xs:simpleType>
463 <xs:complexType name="NetworkLocation" abstract="true">
464   <xs:complexContent>
465     <xs:extension base="D2LogicalModel:Location">
466       <xs:sequence>
467         <xs:element name="networkLocationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
468       </xs:sequence>
469     </xs:extension>
470   </xs:complexContent>
471 </xs:complexType>
472 <xs:simpleType name="NonNegativeInteger">
473   <xs:restriction base="xs:nonNegativeInteger" />
474 </xs:simpleType>
475 <xs:complexType name="OpenlrBaseLocationReferencePoint" abstract="true">
476   <xs:sequence>
477     <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" />
478     <xs:element name="openlrLineAttributes" type=
↪ "D2LogicalModel:OpenlrLineAttributes" />
479     <xs:element name="openlrBaseLocationReferencePointExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
480   </xs:sequence>
481 </xs:complexType>
482 <xs:complexType name="OpenlrBasePointLocation" abstract="true">
483   <xs:sequence>
484     <xs:element name="openlrSideOfRoad" type="D2LogicalModel:OpenlrSideOfRoadEnum"
↪ minOccurs="1" maxOccurs="1" />
485     <xs:element name="openlrOrientation" type="D2LogicalModel:OpenlrOrientationEnum
↪ " minOccurs="1" maxOccurs="1" />
486     <xs:element name="openlrPositiveOffset" type=
↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
487     <xs:element name="openlrLocationReferencePoint" type=
↪ "D2LogicalModel:OpenlrLocationReferencePoint" />
488     <xs:element name="openlrLastLocationReferencePoint" type=
↪ "D2LogicalModel:OpenlrLastLocationReferencePoint" />
489     <xs:element name="openlrBasePointLocationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
490   </xs:sequence>
491 </xs:complexType>
492 <xs:complexType name="OpenlrExtendedLinear">
493   <xs:sequence>
494     <xs:element name="firstDirection" type=
↪ "D2LogicalModel:OpenlrLineLocationReference" />
495     <xs:element name="oppositeDirection" type=
↪ "D2LogicalModel:OpenlrLineLocationReference" minOccurs="0" />
496   </xs:sequence>
497 </xs:complexType>
498 <xs:complexType name="OpenlrExtendedPoint">
499   <xs:sequence>
500     <xs:element name="openlrPointLocationReference" type=
↪ "D2LogicalModel:OpenlrPointLocationReference" />

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501     </xs:sequence>
502 </xs:complexType>
503 <xs:simpleType name="OpenlrFormOfWayEnum">
504     <xs:restriction base="xs:string">
505         <xs:enumeration value="undefined" />
506         <xs:enumeration value="motorway" />
507         <xs:enumeration value="multipleCarriageway" />
508         <xs:enumeration value="singleCarriageway" />
509         <xs:enumeration value="roundabout" />
510         <xs:enumeration value="slipRoad" />
511         <xs:enumeration value="trafficSquare" />
512         <xs:enumeration value="other" />
513     </xs:restriction>
514 </xs:simpleType>
515 <xs:simpleType name="OpenlrFunctionalRoadClassEnum">
516     <xs:restriction base="xs:string">
517         <xs:enumeration value="FRC0" />
518         <xs:enumeration value="FRC1" />
519         <xs:enumeration value="FRC2" />
520         <xs:enumeration value="FRC3" />
521         <xs:enumeration value="FRC4" />
522         <xs:enumeration value="FRC5" />
523         <xs:enumeration value="FRC6" />
524         <xs:enumeration value="FRC7" />
525     </xs:restriction>
526 </xs:simpleType>
527 <xs:complexType name="OpenlrGeoCoordinate">
528     <xs:sequence>
529         <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" />
530         <xs:element name="openlrGeoCoordinateExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
531     </xs:sequence>
532 </xs:complexType>
533 <xs:complexType name="OpenlrLastLocationReferencePoint">
534     <xs:complexContent>
535         <xs:extension base="D2LogicalModel:OpenlrBaseLocationReferencePoint">
536             <xs:sequence>
537                 <xs:element name="openlrLastLocationReferencePointExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
538             </xs:sequence>
539         </xs:extension>
540     </xs:complexContent>
541 </xs:complexType>
542 <xs:complexType name="OpenlrLineAttributes">
543     <xs:sequence>
544         <xs:element name="openlrFunctionalRoadClass" type=
↪ "D2LogicalModel:OpenlrFunctionalRoadClassEnum" minOccurs="1" maxOccurs="1" />
545         <xs:element name="openlrFormOfWay" type="D2LogicalModel:OpenlrFormOfWayEnum"
↪ minOccurs="1" maxOccurs="1" />
546         <xs:element name="openlrBearing" type="D2LogicalModel:AngleInDegrees" minOccurs=
↪ "1" maxOccurs="1" />
547         <xs:element name="openlrLineAttributesExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
548     </xs:sequence>
549 </xs:complexType>
550 <xs:complexType name="OpenlrLineLocationReference">
551     <xs:sequence>

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552     <xs:element name="openlrLocationReferencePoint" type=
↪ "D2LogicalModel:OpenlrLocationReferencePoint" maxOccurs="unbounded" />
553     <xs:element name="openlrLastLocationReferencePoint" type=
↪ "D2LogicalModel:OpenlrLastLocationReferencePoint" />
554     <xs:element name="openlrOffsets" type="D2LogicalModel:OpenlrOffsets" minOccurs=
↪ "0" />
555     <xs:element name="openlrLineLocationReferenceExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
556   </xs:sequence>
557 </xs:complexType>
558 <xs:complexType name="OpenlrLocationReferencePoint">
559   <xs:complexContent>
560     <xs:extension base="D2LogicalModel:OpenlrBaseLocationReferencePoint">
561       <xs:sequence>
562         <xs:element name="openlrPathAttributes" type=
↪ "D2LogicalModel:OpenlrPathAttributes" />
563         <xs:element name="openlrLocationReferencePointExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
564       </xs:sequence>
565     </xs:extension>
566   </xs:complexContent>
567 </xs:complexType>
568 <xs:complexType name="OpenlrOffsets">
569   <xs:sequence>
570     <xs:element name="openlrPositiveOffset" type=
↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
571     <xs:element name="openlrNegativeOffset" type=
↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
572     <xs:element name="openlrOffsetsExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
573   </xs:sequence>
574 </xs:complexType>
575 <xs:simpleType name="OpenlrOrientationEnum">
576   <xs:restriction base="xs:string">
577     <xs:enumeration value="noOrientationOrUnknown" />
578     <xs:enumeration value="withLineDirection" />
579     <xs:enumeration value="againstLineDirection" />
580     <xs:enumeration value="both" />
581   </xs:restriction>
582 </xs:simpleType>
583 <xs:complexType name="OpenlrPathAttributes">
584   <xs:sequence>
585     <xs:element name="openlrLowestFRCToNextLRPoint" type=
↪ "D2LogicalModel:OpenlrFunctionalRoadClassEnum" minOccurs="1" maxOccurs="1" />
586     <xs:element name="openlrDistanceToNextLRPoint" type=
↪ "D2LogicalModel:NonNegativeInteger" minOccurs="1" maxOccurs="1" />
587     <xs:element name="openlrPathAttributesExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
588   </xs:sequence>
589 </xs:complexType>
590 <xs:complexType name="OpenlrPointAlongLine">
591   <xs:complexContent>
592     <xs:extension base="D2LogicalModel:OpenlrBasePointLocation">
593       <xs:sequence>
594         <xs:element name="openlrPointAlongLineExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />

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595     </xs:sequence>
596   </xs:extension>
597 </xs:complexContent>
598 </xs:complexType>
599 <xs:complexType name="OpenlrPointLocationReference">
600   <xs:sequence>
601     <xs:element name="openlrGeoCoordinate" type="D2LogicalModel:OpenlrGeoCoordinate
602     ↪" minOccurs="0" />
603     <xs:element name="openlrPoiWithAccessPoint" type=
604     ↪"D2LogicalModel:OpenlrPoiWithAccessPoint" minOccurs="0" />
605     <xs:element name="openlrPointAlongLine" type=
606     ↪"D2LogicalModel:OpenlrPointAlongLine" minOccurs="0" />
607     <xs:element name="openlrPointLocationReferenceExtension" type="D2LogicalModel:_
608     ↪ExtensionType" minOccurs="0" />
609   </xs:sequence>
610 </xs:complexType>
611 <xs:complexType name="OpenlrPoiWithAccessPoint">
612   <xs:complexContent>
613     <xs:extension base="D2LogicalModel:OpenlrBasePointLocation">
614       <xs:sequence>
615         <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" /
616         ↪>
617         <xs:element name="openlrPoiWithAccessPointExtension" type="D2LogicalModel:_
618         ↪ExtensionType" minOccurs="0" />
619       </xs:sequence>
620     </xs:extension>
621 </xs:complexContent>
622 </xs:complexType>
623 <xs:simpleType name="OpenlrSideOfRoadEnum">
624   <xs:restriction base="xs:string">
625     <xs:enumeration value="onRoadOrUnknown" />
626     <xs:enumeration value="right" />
627     <xs:enumeration value="left" />
628     <xs:enumeration value="both" />
629   </xs:restriction>
630 </xs:simpleType>
631 <xs:complexType name="PayloadPublication" abstract="true">
632   <xs:sequence>
633     <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0" ↪
634     ↪maxOccurs="1" />
635     <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs="1" ↪
636     ↪maxOccurs="1" />
637     <xs:element name="publicationCreator" type=
638     ↪"D2LogicalModel:InternationalIdentifier" />
639     <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_
640     ↪ExtensionType" minOccurs="0" />
641   </xs:sequence>
642   <xs:attribute name="lang" type="D2LogicalModel:Language" use="required" />
643 </xs:complexType>
644 <xs:simpleType name="Percentage">
645   <xs:restriction base="D2LogicalModel:Float" />
646 </xs:simpleType>
647 <xs:complexType name="PercentageDistanceAlongLinearElement">
648   <xs:complexContent>
649     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
650       <xs:sequence>

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641     <xs:element name="percentageDistanceAlong" type="D2LogicalModel:Percentage"
↪ minOccurs="1" maxOccurs="1" />
642     <xs:element name="percentageDistanceAlongLinearElementExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
643     </xs:sequence>
644     </xs:extension>
645     </xs:complexContent>
646   </xs:complexType>
647   <xs:complexType name="Point">
648     <xs:complexContent>
649       <xs:extension base="D2LogicalModel:NetworkLocation">
650         <xs:sequence>
651           <xs:element name="alertCPoint" type="D2LogicalModel:AlertCPoint" minOccurs=
↪ "0" />
652           <xs:element name="pointAlongLinearElement" type=
↪ "D2LogicalModel:PointAlongLinearElement" minOccurs="0" />
653           <xs:element name="pointByCoordinates" type=
↪ "D2LogicalModel:PointByCoordinates" minOccurs="0" />
654           <xs:element name="pointExtension" type="D2LogicalModel:_PointExtensionType"
↪ minOccurs="0" />
655         </xs:sequence>
656       </xs:extension>
657     </xs:complexContent>
658   </xs:complexType>
659   <xs:complexType name="PointAlongLinearElement">
660     <xs:sequence>
661       <xs:element name="administrativeAreaOfPoint" type=
↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
662       <xs:element name="directionBoundAtPoint" type="D2LogicalModel:DirectionEnum"
↪ minOccurs="0" maxOccurs="1" />
663       <xs:element name="directionRelativeAtPoint" type=
↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="0" maxOccurs="1" />
664       <xs:element name="linearElement" type="D2LogicalModel:LinearElement" />
665       <xs:element name="distanceAlongLinearElement" type=
↪ "D2LogicalModel:DistanceAlongLinearElement" />
666       <xs:element name="pointAlongLinearElementExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
667     </xs:sequence>
668   </xs:complexType>
669   <xs:complexType name="PointByCoordinates">
670     <xs:sequence>
671       <xs:element name="bearing" type="D2LogicalModel:NonNegativeInteger" minOccurs="0
↪ maxOccurs="1" />
672       <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates" />
673       <xs:element name="pointByCoordinatesExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
674     </xs:sequence>
675   </xs:complexType>
676   <xs:complexType name="PointByMultiCoordinates">
677     <xs:sequence>
678       <xs:element name="sjtskPointCoordinates" type=
↪ "D2LogicalModel:SJTSKPointCoordinates" minOccurs="0" />
679       <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates"
↪ minOccurs="0" />
680     </xs:sequence>
681   </xs:complexType>

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682 <xs:complexType name="PointCoordinates">
683   <xs:sequence>
684     <xs:element name="latitude" type="D2LogicalModel:Float" minOccurs="1" maxOccurs=
↪ "1" />
685     <xs:element name="longitude" type="D2LogicalModel:Float" minOccurs="1"
↪ maxOccurs="1" />
686     <xs:element name="pointCoordinatesExtension" type="D2LogicalModel:_ExtensionType
↪ " minOccurs="0" />
687   </xs:sequence>
688 </xs:complexType>
689 <xs:complexType name="PointWithinLinearGNElement">
690   <xs:sequence>
691     <xs:element name="sectionId" type="D2LogicalModel:NonNegativeInteger" minOccurs=
↪ "0" maxOccurs="1" />
692     <xs:element name="directionRelativeOnLinearSection" type=
↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="1" maxOccurs="1" />
693     <xs:element name="networkVersion" type="D2LogicalModel:String" minOccurs="1"
↪ maxOccurs="1" />
694     <xs:element name="distanceAlongLinearElement" type=
↪ "D2LogicalModel:DistanceAlongLinearElement" />
695   </xs:sequence>
696 </xs:complexType>
697 <xs:complexType name="PredefinedLocation">
698   <xs:complexContent>
699     <xs:extension base="D2LogicalModel:PredefinedLocationContainer">
700       <xs:sequence>
701         <xs:element name="predefinedLocationName" type=
↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
702         <xs:element name="location" type="D2LogicalModel:Location" />
703         <xs:element name="predefinedLocationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
704       </xs:sequence>
705       <xs:attribute name="id" type="xs:string" use="required" />
706       <xs:attribute name="version" type="xs:string" use="required" />
707     </xs:extension>
708   </xs:complexContent>
709 </xs:complexType>
710 <xs:complexType name="PredefinedLocationContainer" abstract="true">
711   <xs:sequence>
712     <xs:element name="predefinedLocationContainerExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
713   </xs:sequence>
714 </xs:complexType>
715 <xs:complexType name="PredefinedLocationsPublication">
716   <xs:complexContent>
717     <xs:extension base="D2LogicalModel:PayloadPublication">
718       <xs:sequence>
719         <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation
↪ " />
720         <xs:element name="predefinedLocationContainer" type=
↪ "D2LogicalModel:PredefinedLocationContainer" maxOccurs="unbounded" />
721         <xs:element name="predefinedLocationsPublicationExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
722       </xs:sequence>
723     </xs:extension>
724   </xs:complexContent>

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725 </xs:complexType>
726 <xs:complexType name="PredefinedNonOrderedLocationGroup">
727   <xs:complexContent>
728     <xs:extension base="D2LogicalModel:PredefinedLocationContainer">
729       <xs:sequence>
730         <xs:element name="predefinedNonOrderedLocationGroupName" type=
731 ↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
732         <xs:element name="predefinedLocation" type=
733 ↪ "D2LogicalModel:PredefinedLocation" minOccurs="2" maxOccurs="unbounded" />
734         <xs:element name="predefinedNonOrderedLocationGroupExtension" type=
735 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
736       </xs:sequence>
737       <xs:attribute name="id" type="xs:string" use="required" />
738       <xs:attribute name="version" type="xs:string" use="required" />
739     </xs:extension>
740   </xs:complexContent>
741 </xs:complexType>
742 <xs:complexType name="Referent">
743   <xs:sequence>
744     <xs:element name="referentIdentifier" type="D2LogicalModel:String" minOccurs="1
745 ↪ " maxOccurs="1" />
746     <xs:element name="referentName" type="D2LogicalModel:String" minOccurs="0"
747 ↪ maxOccurs="1" />
748     <xs:element name="referentType" type="D2LogicalModel:ReferentTypeEnum"
749 ↪ minOccurs="1" maxOccurs="1" />
750     <xs:element name="referentDescription" type="D2LogicalModel:MultilingualString"
751 ↪ minOccurs="0" maxOccurs="1" />
752     <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates"
753 ↪ minOccurs="0" />
754     <xs:element name="referentExtension" type="D2LogicalModel:_ExtensionType"
755 ↪ minOccurs="0" />
756   </xs:sequence>
757 </xs:complexType>
758 <xs:simpleType name="ReferentTypeEnum">
759   <xs:restriction base="xs:string">
760     <xs:enumeration value="boundary" />
761     <xs:enumeration value="intersection" />
762     <xs:enumeration value="referenceMarker" />
763     <xs:enumeration value="landmark" />
764     <xs:enumeration value="roadNode" />
765   </xs:restriction>
766 </xs:simpleType>
767 <xs:complexType name="SJTSKPointCoordinates">
768   <xs:sequence>
769     <xs:element name="sjtskX" type="D2LogicalModel:Float" minOccurs="1" maxOccurs="1
770 ↪ " />
771     <xs:element name="sjtskY" type="D2LogicalModel:Float" minOccurs="1" maxOccurs="1
772 ↪ " />
773     <xs:element name="sjtskPointCoordinatesExtension" type="D2LogicalModel:_
774 ↪ ExtensionType" minOccurs="0" />
775   </xs:sequence>
776 </xs:complexType>
777 <xs:simpleType name="String">
778   <xs:restriction base="xs:string">
779     <xs:maxLength value="1024" />
780   </xs:restriction>

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769 </xs:simpleType>
770 <xs:simpleType name="UrgencyEnum">
771   <xs:restriction base="xs:string">
772     <xs:enumeration value="extremelyUrgent" />
773     <xs:enumeration value="urgent" />
774     <xs:enumeration value="normalUrgency" />
775   </xs:restriction>
776 </xs:simpleType>
777 <xs:complexType name="VersionedReference">
778   <xs:attribute name="id" type="xs:string" use="required" />
779   <xs:attribute name="version" type="xs:string" use="required" />
780 </xs:complexType>
781 </xs:schema>
```

CHANGELOG

version 1.2

- corrected location of globalNetworkLinear and globalNetworkPoint in the XML tree (as extension)
- corrected type of the networkVersion to the simple string from multilingual string
- excluded from documentation not used location reference methods (mainly point location references since they are not used for predefined location sets)
- simplified documentation
- included changelog in the documentation

version 1.1.0

- added pointByCoordinates to concepts, schema and samples
- added feed type to schema and samples
- added OpenLR location to schema

version 1.0.1

- revision of the documentation, proofreading, changes to examples

version 1.0.0

- Initial format and documentation

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- [d2supp] DATEX II Current version/Supporting, <https://docs.datex2.eu/>.

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